Lepton Generations and the Missing Mass Operator in Deterministic Field Closure

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Abstract

The three-order mass law of Paper I [Smart, 2026a] recovers the proton-to-electron mass ratio at the 2×10^{-4} level but cannot accommodate heavier lepton masses by shell-support extension alone. We prove a formal no-go for contiguous boundary-starting shell supports: for supports of the form $S = \{1, 2, \ldots, k\}$, the combinatorial invariant grows factorially and no such support yields mass ratios in the muon or tau range. This identifies the need for an additional mass-lifting mechanism independent of shell-support size.

We then formulate a conditional winding ansatz for boundary excitations,

$$f(w) = \varphi^5 (w - 1)^{1/\varphi},$$

which leaves the proton-to-electron result unchanged because $f(1) = 0$. Within the present framework, the exponent $1/\varphi$ follows if the relevant effective spectral dimension is identified with $d_s = \varphi$. That identification is not derived in this paper; it is motivated numerically by the separate F4 Fibonacci hierarchical construction analysed in Paper III [Smart, 2026b]. Under this conjectural spectral identification, the winding operator reproduces the muon-to-electron ratio to 0.5% and the tau-to-electron ratio to 3.7% with zero fitted continuous parameters.

This paper therefore establishes a derived obstruction, proposes a conditional operator that overcomes it, and makes explicit which parts of the lepton-generation programme remain conjectural.

1 Introduction

Paper I [Smart, 2026a] establishes a three-order mass law for the baryon-lepton hierarchy within the deterministic closure framework, recovering the proton-to-electron mass ratio at the $\sim 2 \times 10^{-4}$ level with zero fitted continuous parameters. That law, however, cannot reproduce heavier lepton masses. The present paper shows that this limitation is structural, provable as a no-go theorem (Section 2), and identifies the nature of the missing mechanism.

Within the present framework, the required additional mechanism is a spectral layer supporting a winding-dependent mass contribution. Under the shell-support graph used in the base formalism, the effective spectral dimension is path-like ($d_s = 1$), not the value $d_s = \varphi$ required by the winding ansatz.

The present paper has a narrower scope than a full theory of lepton masses. Its main derived result is a no-go theorem for contiguous shell-support extension. Its main positive result is conditional: if the relevant effective spectral dimension is identified with φ, then a winding ansatz yields the muon and tau mass ratios with the accuracies reported below. The spectral support for that identification is treated here only in summary form; detailed discussion of the 600-cell eigenvalue correspondence, the F4 hierarchical construction, and the retraction of the earlier 600-cell $d_s^{\text{eff}} \approx \varphi$ claim belongs to Paper III [Smart, 2026b].

The paper is organised as follows. Section 2 proves the shell-extension no-go theorem. Section 3 identifies the resulting spectral obstruction. Sections 4 and 5 introduce the conditional winding operator and its lepton-mass predictions. Section 7 summarises the external spectral motivation from Paper III. Section 8 gives the logical dependency chain, and Section 9 classifies the epistemic status of each claim.

Epistemic status and scope

This paper has three layers of claim:

1. **Derived result.** The shell-extension no-go theorem and the resulting spectral obstruction are derived within the Paper I framework.

2. **Conditional construction.** The winding operator is proposed as a conditional extension, valid if the relevant effective spectral dimension is identified with $d_s = \varphi$.

3. **External support.** The 600-cell eigenvalue correspondence and the F4 hierarchical construction are used only as external supporting motivation, following Paper III [Smart, 2026b].

The paper does not claim an analytic derivation of $d_s = \varphi$, nor a first-principles derivation of the winding operator.

2 No-Go Theorem for Shell-Extension Leptons

Status: derived.

Proposition 1 (No-Go for Contiguous Boundary-Starting Shell Extensions)**.** *Under the three-order mass law of [Smart, 2026a] with assignment rules R1–R5, no boundary-starting shell support $S = \{1, 2, \ldots, k\}$ produces a mass ratio $m(S)/m_e$ in the interval* $[100, 4000]$.

Proof. For $S = \{1, \ldots, k\}$, the combinatorial invariant is $C(S) = k! - k(k+1)/2 - 1$. The three-order exponent relative to the electron is:

$$\Delta\mathcal{E}(k) = [C(\{1, \ldots, k\}) - C(\{1\})] + \Delta g_1 + \Delta g_2$$

Direct computation gives:

The function $k! - k(k+1)/2 - 1$ grows super-exponentially: C jumps from -1 (at $k = 3$) to 13 (at $k = 4$), producing a ratio gap from ~ 1 to ~ 1200 with no intermediate values. The muon (~ 207) and tau (~ 3477) mass ratios lie in this gap. No integer k reaches the interval $[100, 4000]$. □ □

Table 1: Mass ratios under shell-support extension from boundary.

k	C	$\Delta\mathcal{E}$	Computed ratio	Observed target
1	-1	0	1.00	(electron)
2	-2	-0.50	0.79	muon: 206.8
3	-1	$+0.62$	1.35	tau: 3477
4	13	$+14.70$	1182	—
5	104	$+105.76$	$> 10^{21}$	—

Corollary 2. *Lepton generations require a mass-lifting mechanism independent of shell-support size.*

This derived obstruction is the central result of the present paper. Everything that follows is an attempt to characterise the type of mechanism required to overcome it.

3 The Spectral Gap

Status: derived obstruction.

The no-go theorem constrains the missing mechanism. It must:

1. generate mass ratios of $\sim 10^2$–10^3 for boundary leptons,
2. not modify the proton-to-electron ratio (which uses winding $w = 1$ for both particles),
3. be discrete (to preserve the discrete particle spectrum),
4. have a structural origin within the closure framework.

Within the present rule set, the only remaining discrete degree of freedom available for a boundary lepton (shell $\{1\}$, fixed by R1–R2) is the *winding number* w. A winding-dependent boundary excitation operator $f(w)$ with $f(1) = 0$ satisfies constraint (2) by construction.

Within the present winding ansatz, the w-th excitation is taken to scale as $(w-1)^{d/(d+1)}$ in an effective dimension d. Recovering the muon mass motivates the condition:

$$\frac{d}{d+1} = \frac{1}{\varphi}$$

This equation has the unique solution $d = \varphi$, via the defining property of the golden ratio: $\varphi/(\varphi + 1) = 1/\varphi$. The identity is exact.

The *obstruction* is that under the shell-support graph used in the base formalism, the effective spectral dimension is path-like and evaluates to $d_s = 1$, not to $d_s = \varphi$. The exponent $1/\varphi$ is therefore available only conditionally: it follows rigorously if $d_s = \varphi$, but that spectral identification must come from outside the base shell structure.

The point is therefore not that $d_s = \varphi$ has been established here, but that the winding programme requires some spectral ingredient outside the base shell-support graph. The present paper isolates that dependency explicitly.

4 The Winding Excitation Operator

4.1 Operator form

The winding contribution to the mass exponent is:

$$f(w) = \varphi^N \cdot (w-1)^{1/\varphi} \tag{1}$$

where N is the manifold shell count and $\varphi = (1+\sqrt{5})/2$. This satisfies $f(1) = 0$ by construction.

4.2 Status of the exponent $1/\varphi$

Status: conditional. The exponent $1/\varphi$ follows within the present spectral-scaling ansatz if the relevant effective spectral dimension is identified with $d_s = \varphi$.

The identity $\varphi/(\varphi+1) = 1/\varphi$ is exact (it is the defining property of the golden ratio). The exponent is therefore conditionally derived: it follows rigorously from the spectral-dimension identification, which is itself conjectured.

4.3 Status of the coefficient φ^N

Status: structurally selected.

Within the restricted family φ^k with integer k, the choice $k = 5$ is the only member within 1% of the coefficient required by the muon mass:

Table 2: Coefficient candidates in the φ^k family.

k	φ^k	Muon error	Status
4	6.854	38.1%	Too small
5	**11.090**	**0.097%**	**Only near-match in tested φ^k family**
6	17.944	62.0%	Too large

Within the framework, $N = 5$ is interpreted as the total manifold shell count, giving a structural reading of the coefficient. This is an interpretation, not an independent derivation.

Status of $N = 5$: postulated. The shell count is an input of the framework, not derived from it.

5 Mass Predictions

Lepton generations are reinterpreted as winding excitations of the boundary closure class:

$$\text{electron:} \quad \{1\},\ w = 1 \quad \text{(ground state)} \tag{2}$$

$$\text{muon:} \quad \{1\},\ w = 2 \quad \text{(first excitation)} \tag{3}$$

$$\text{tau:} \quad \{1\},\ w = 3 \quad \text{(second excitation)} \tag{4}$$

With $f(w) = \varphi^5(w-1)^{1/\varphi}$ and zero fitted continuous parameters:

The muon value uses $f(2) = \varphi^5$. The tau value is a genuine test: $f(3) = \varphi^5 \cdot 2^{1/\varphi}$ is determined entirely by the operator form, not calibrated to the tau.

The proton-to-electron ratio is unaffected: both particles have $w = 1$, so $f(1) = 0$. The Paper I result $m_p/m_e = \varphi^{1265/81} \approx 1835.8$ is unchanged.

Table 3: Lepton mass ratios under the winding operator. These results are conditional on the conjectural spectral identification $d_s = \varphi$.

Particle	Winding	Computed ratio	Observed	Agreement
Electron	$w = 1$	1 (reference)	—	—
Muon	$w = 2$	$\varphi^{\varphi^5} \approx 207.8$	206.8	0.5%
Tau	$w = 3$	$\varphi^{\varphi^5 \cdot 2^{1/\varphi}} \approx 3607$	3477	3.7%

6 Extended Factorised Law

The extended mass law, conditional on the winding ansatz and incorporating the Paper I baryon-lepton law, takes the factorised form:

$$m(\mathcal{A}) = \Lambda \cdot \varphi^{C(\mathcal{A})} \cdot \varphi^{|E|/|V|} \cdot \varphi^{-\mathrm{Var}(\deg)\,|E|/|V|^2} \cdot \varphi^{f(w)} \tag{5}$$

where $|E|$ and $|V|$ are the edge and vertex counts of the closure graph, and $f(w)$ is defined by Eq. (1). In the present framework, each factor is assigned to a distinct structural sector and the leading-order law is taken to factorise across those sectors. For particles with $w = 1$ (electron, proton, neutron), the winding factor equals $\varphi^0 = 1$ and the law reduces to Paper I.

7 External Spectral Motivation

The winding operator requires a spectral ingredient ($d_s = \varphi$) not available from the base shell-support graph. This section summarises two external spectral results treated in detail in Paper III [Smart, 2026b].

7.1 600-cell eigenvalue correspondence

Paper III [Smart, 2026b] establishes an exact computed correspondence between the closure-class mass programme and the 600-cell spectrum. The graph Laplacian of the 600-cell vertex graph contains the eigenvalue $\lambda = 15$ with multiplicity 16, while the closure invariant difference between proton and electron is $\Delta C = 15$. Thus the leading-order mass exponent is equal to an eigenvalue of the polytope Laplacian.

This is not used here as a derivation of the winding operator. Its role in the present paper is supportive: it shows that the closure-class mass programme already has a non-trivial exact spectral correspondence at leading order.

7.2 F4 numerical motivation

Paper III [Smart, 2026b] studies a separate F4 Fibonacci hierarchical extension of the φ-manifold. This is not the 600-cell vertex graph, but a constructed hierarchical family introduced to test whether φ-structured shell organisation can produce effective spectral dimensions approaching φ under controlled scaling.

The role of the F4 construction in the present paper is limited: it provides exploratory numerical motivation for identifying the relevant effective spectral dimension with φ in the winding ansatz.

Key results from [Smart, 2026b]:

- The effective spectral dimension of the F4 family converges from above toward φ as the base size b increases: $d_s^{\mathrm{eff}\,HK} = 1.635$ at $V = 4960$.
- Empirical $1/V$ extrapolation gives $d_s(\infty) = 1.628$ ($R^2 = 0.996$).
- Among all tested control families, F4 is the only one whose effective spectral dimension trends toward φ.
- The random-walk estimate is robust to 20% edge-weight perturbation ($\Delta d_s^{\mathrm{eff}} \approx 0.001$).

These results provide exploratory numerical support for the conjectural identification $d_s = \varphi$ in the extended hierarchical setting. Detailed analysis, controls, and the explicit retraction of the earlier 600-cell $d_s^{\mathrm{eff}} \approx \varphi$ claim belong to Paper III [Smart, 2026b].

8 Dependency Chain

The logical structure of the lepton generation programme is:

$$
\begin{array}{lcl}
\text{F4 Fibonacci construction} & \xrightarrow{\text{numerical convergence}} & d_s^{\mathrm{eff}} \to \varphi \\
d_s^{\mathrm{eff}} \approx \varphi & \xrightarrow{\text{conjectural identification}} & d_s = \varphi \\
d_s = \varphi & \xrightarrow{\text{exact identity}} & \text{exponent} = 1/\varphi \\
\text{exponent } +\varphi^5 \text{ coefficient} & \xrightarrow{\text{winding operator}} & f(w) = \varphi^5(w-1)^{1/\varphi} \\
f(w) & \xrightarrow{\text{evaluation}} & \text{lepton mass ratios}
\end{array}
$$

Independent support: exact 600-cell eigenvalue correspondence $\lambda = 15 = \Delta C$, as summarised from Pap

Each arrow carries a stated epistemic status. The chain is as strong as its weakest link: the conjectural identification $d_s = \varphi$.

9 Claim Status

What this paper does not claim

This paper does not claim an analytic proof that $d_s = \varphi$, nor that the winding operator is fully derived from first principles. It does not derive the shell count $N = 5$ or the coefficient φ^5. It does not claim that the 600-cell vertex graph itself has effective spectral dimension $d_s^{\mathrm{eff}} \approx \varphi$; that earlier claim has been retracted in Paper III [Smart, 2026b]. In the present paper, the 600-cell contributes only an exact eigenvalue correspondence, while the F4 construction provides exploratory numerical support for the conjectural spectral identification.

The paper's main claims are: a derived shell-extension no-go theorem, a derived spectral obstruction within the base shell-support graph, and a conditional winding operator supported by external spectral motivation.

Table 4: Claim status table.

Claim	Status	Basis
Shell-extension no-go	DERIVED	Exhaustive computation
Spectral gap obstruction	DERIVED	Base shell-support graph g
$\lambda = 15$ occurs in the 600-cell spectrum	EXACT COMPUTATION	Summarised from Paper II
$d_s = \varphi$ in the winding construction	CONJECTURED	Motivated numerically by
Exponent $1/\varphi$	CONDITIONAL	Requires $d_s = \varphi$
Coefficient φ^5	STRUCTURALLY SELECTED	Unique in tested φ^k family
Shell count $N = 5$	POSTULATED	Framework input
Full winding operator	CONDITIONAL	Requires spectral identifica
Muon ratio (0.5%)	CONDITIONAL	Follows from operator
Tau ratio (3.7%)	CONDITIONAL	Follows from operator; not
F4 effective spectral dimensions approach φ	NUMERICAL OBSERVATION	Summarised from Paper II

10 Open Problems

1. **Exact proof of $d_s = \varphi$.** The F4 extrapolation provides exploratory numerical motivation for $d_s = \varphi$ in the extended hierarchical construction, but does not prove it. An analytical derivation — from the spectral zeta function, renormalisation group, or heat-kernel asymptotics — remains the primary formal target.

2. **Coefficient derivation.** The coefficient φ^5 is structurally selected (unique in its family) but not derived from the closure geometry. Whether the F4 spectral recursion or the 600-cell structure provides an independent route to φ^5 is unknown.

3. **Canonical geometric selection.** Why the 600-cell should play a privileged role in the closure framework remains open. At present its strongest support is the exact eigenvalue correspondence $\lambda = 15 = \Delta C$, not a spectral-dimension identification.

4. **Tau mass correction.** The tau ratio has 3.7% structural error. Whether a higher-order winding correction, analogous to the graph corrections in Paper I, can reduce this is unknown.

5. **Neutron-proton splitting.** A symmetry-sector correction φ^{δ_σ} with $\delta_\sigma \approx 0.003$ is compatible with the architecture but not derived.

6. **Extension to quarks and heavier hadrons.**

11 Discussion

The central structural result is the no-go theorem: shell-support extension cannot produce lepton generations. This identifies the class of the missing mechanism (spectral, winding-dependent) without determining it uniquely.

The 600-cell contributes to the lepton-generation programme only indirectly in this paper, through the exact eigenvalue correspondence $\lambda = 15 = \Delta C$ established separately in Paper III. The F4 Fibonacci hierarchical construction provides exploratory numerical motivation that a φ-structured geometry can produce effective spectral dimensions approaching φ. These two spectral inputs are distinct: the former is an exact finite-graph

correspondence, while the latter is a separate numerical extension. The present paper does not derive the winding operator from either of them; it uses them only as external support for the conditional programme.

The winding operator $f(w) = \varphi^5(w-1)^{1/\varphi}$, conditional on the spectral identification, recovers three charged lepton masses from a single functional form with zero fitted continuous parameters. The agreement is structural rather than metrological: the muon and tau masses are known to much higher precision than 0.5% and 3.7% respectively.

Whether the spectral identification $d_s = \varphi$ can be established analytically is the primary remaining formal question. If it can, the exponent $1/\varphi$ becomes derived rather than conditional, upgrading the status of the entire lepton programme.

12 Conclusion

A formal no-go theorem establishes that lepton generations cannot arise from shell-support extension within the closure-class mass law. The no-go theorem identifies the need for a winding-dependent boundary excitation operator, and the form $f(w) = \varphi^5(w-1)^{1/\varphi}$ is proposed as the minimal structurally compatible candidate within the present framework. The exponent $1/\varphi$ follows within the spectral-scaling ansatz under the identification $d_s = \varphi$; this identification is motivated numerically by the F4 Fibonacci hierarchical construction ($d_s^{\text{eff}} = 1.635$ at $V = 4960$, extrapolating to 1.628), but has not been analytically proven.

The operator recovers the muon-to-electron ratio to 0.5% and the tau-to-electron ratio to 3.7% with zero fitted continuous parameters. Together with the proton-to-electron law of Paper I [Smart, 2026a], the framework provides a constrained, conditional structural extension from the baryon-lepton hierarchy to the three charged leptons.

The programme's epistemic structure is explicit: the no-go theorem and the shell-support obstruction are derived; the 600-cell contributes an exact eigenvalue correspondence $\lambda = 15 = \Delta C$; the identification $d_s = \varphi$ remains conjectural; and the winding operator and its lepton-mass predictions are conditional on that conjecture. The F4 hierarchical construction provides exploratory numerical motivation for the spectral identification, but not an analytic proof. Closing that gap remains the primary formal target.

References

Lee Smart. Proton-to-electron mass ratio from closure geometry and graph invariants. 2026a. Companion paper. Preprint, Institute of Vibrational Field Dynamics.

Lee Smart. A fibonacci hierarchical extension of the φ-manifold and its effective spectral dimension. 2026b. Companion paper. Preprint, Institute of Vibrational Field Dynamics.